ECOMATCH AI: AN AUTOMATED PRODUCT INQUIRY RESPONSE SYSTEM FOR LIGHT RECYCLING PROGRAM

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BACKGROUND & PURPOSE

- **Product Care Association (PCA)**
  - Recycling post-consumer lighting products across Canada
- **Manual product inquiry process**
  - Inefficiencies and delays
  - Inconsistent decision-making

RESEARCH QUESTION

- **How can I design and implement a product**
  - ✓ End-to-end
  - ✓ AI-powered
  - ✓ Integrates text and image data
  - ✓ Minimizes response times
  - ✓ Enhances matching accuracy

NOVELTY

- **PCA's first ML-powered system**

Methodology and Tech Stack

Web Portal

- **Frontend:** Python Flask and Bootstrap
- **Backend:** Python Flask
- **Data Storage:** Azure Table Storage and Azure Blob Storage

Text Similarity

- **Technique:** Sentence-BERT (SBERT) – fine-tuned transformer for semantic similarity
  - **Short text:** all-mpnet-base-v2
  - **Long text:** bert-large-nli-stsb-mean-tokens

Image Similarity

- **Technique:** Convolutional Neural Networks (CNN) based feature extraction for image similarity
  - **Best performer:** ResNet50

Challenges Faced

- CNN model - ResNet50 was slow, taking over 5 minutes to run
- Limited experience using Python in a full-stack development context
- Understanding and applying complex NLP and CNN models
- Insufficient labeled data for training ML models

Lessons Learned

- Precomputing embeddings drastically cut runtime from 5 minutes to 5 seconds, showing the value of optimizing for production
- Persistence and leveraging online resources made it possible to understand complex concepts
- Adding an admin review process improved data quality and helped generate valuable training data over time

Thank You!!!